

American Rocketry Challenge (ARC) 2025 Final Report

E-Town Rocket Bureau

Summary

Starting in August 2024, the E-Town Rocket Bureau (ERB) began designing and building a model rocket for entry into ARC 2025. The rocket, named *Wildkit SRB MK3*, took its name from the school mascot (*Wildkit*), its solid propulsion system (*SRB* for Solid Rocket Booster), and the fact that it was the third major design iteration (*MK3*).

Although initial testing showed promise, the first three versions of the Wildkit SRB were ultimately unsuccessful. However, the ERB team gained significant technical experience and insight throughout the process.

This marked Evanston Township High School's first serious attempt to participate in ARC, building on a previous, but unsuccessful, effort by the Northern Illinois Rocketry Association (NIRA) to launch the program at ETHS in 2018.

Team Members

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Flight Performance and Analysis

Flight #1 – carrying a crew of two eggs – demonstrated visibly lower acceleration compared to other competing rockets. Wildkit SRB MK3 successfully cleared the launch rail but weathercocked into a suboptimal trajectory. The rocket reached an apogee of 450 ft, significantly below the target of 790 ft. Stage separation occurred shortly after apogee as intended. While the booster stage's parachute deployed successfully, the crew stage parachute failed to deploy. This resulted in a high-velocity impact of the crew stage, which suffered catastrophic structural failure. Upon inspection, the ERB team found both eggs destroyed, and the lower, glued fuselage had detached. The booster stage remained largely intact, sustaining only minor burn marks between the upper motor retainer tube and the transition.

The ERB team determined that the crew stage parachute failed to deploy due to suboptimal placement. It had been stored within the crew stage, unlike the booster parachute, which was housed in the transition section. As a result, the ejection charge did not generate enough force to extract the crew parachute, leading to a failure in deployment.

The loss of the egg crew was attributed to two key factors:

- I. The high impact velocity of the crew stage.
- II. The internal carbon fiber rails, which caused damage to the eggs upon repeated impacts during flight.

The below-target apogee of Wildkit SRB MK3 was attributed to three contributing factors:

- I. The rocket's initial mass of 650g was too high to reach the target altitude.
- II. A possible shift of the center of gravity (C_g) behind the center of pressure (C_p) during motor burn may have introduced instability and oscillation, increasing drag.

- III. The fins were undersized, increasing susceptibility to weathercocking when compared to competitor rockets.

Flight #2 incorporated key corrections. The parachute issue was resolved by relocating the crew stage parachute to the transition section alongside the booster parachute. Although apogee data was not recorded due to the altimeter being left off, visual estimation placed the peak at approximately 550 ft. Performance improved due to the removal of the mass adjustment system, the carbon fiber tubes near the nose, and the absence of eggs.

The ERB team believes Flight #2 could have achieved a higher apogee with better C_g/C_p calibration and larger fins. Additionally, the team recommends implementing a mandatory pre-flight checklist to prevent oversights such as failure to activate the altimeter.

Despite the improvements, Wildkit SRB MK3 again launched at a significant angle, although less severe than in Flight #1. This led to a water landing in a nearby reservoir. Both stages remained buoyant and were eventually recovered, but the crew stage suffered severe water damage and was deemed unfit for further flight.

All subsequent flights of Wildkit SRB MK3 were cancelled due to irreparable water damage sustained by the crew stage.

In both flights, the rocket exhibited rapid pitching as it approached apogee. Although this behavior was predicted by [*EulerSim_{BETA}*](#) (see Program Analysis), actual pitch data was not collected, making it impossible to quantitatively verify the simulation's accuracy.

Program Analysis

Development of the Wildkit SRB program progressed slowly. From August through January, the ERB team adhered to a digital prototyping philosophy. While this approach minimized development costs, it proved slow and mentally unengaging. Delays were exacerbated by the team's limited experience with CAD and related software. By the time the team transitioned to traditional prototyping in February, there was insufficient time remaining to order parts and construct the rocket.

Due to late ordering, several critical components—most notably the Aerotech motor and lower body tubes—were not delivered in time. As a result, ERB was forced to make significant design changes, including removing the airbrake system and replacing it with an imprecise initial mass-adjustment method. Compounding these issues, the absence of a centralized system for tracking component orders and deliveries led to critical parts being omitted entirely, forcing last-minute improvisation.

Due to time constraints and limited funding, the team was unable to procure replacement parts or construct a backup rocket. This also led to the removal of the avionics and data collection systems, which had not been fully assembled. On flight day, ERB was unable to replace a damaged parachute due to a lack of spare parts and thus failed to complete qualification flights.

The ERB team concludes that slightly increased funding, coupled with the adoption of a rapid prototyping methodology and improved logistics tracking, would address many of the challenges encountered during this iteration of the Wildkit SRB program.

Recommendations

ERB recommends the following changes, adaptations, inclusions, and continuations to the Wildkit SRB program and, where applicable, to broader rocketry initiatives:

Timeline

- I. A significant portion of the program—including design, prototyping, and testing—should be completed during the summer months, as academic commitments during the school year limit available working time.
- II. Look into joining NIRA after June 30th.

Budget

- III. An increased budget of at least \$600 is recommended. This would reduce design constraints and provide the flexibility to purchase extra components for repairs or additional rockets, enhancing program resilience.

Component Acquisition

- IV. Following recommendations from NIRA, all major components should be ordered during the summer months, when demand is low.
- V. Sufficient components should be procured to build two rockets, along with extra parts for repairs.
- VI. A centralized inventory system should be used to track ordered and received components.

Design Process and Considerations

- VII. Adopt a rapid prototyping workflow:
 - A. Begin with a rough digital design, using numerical estimation methods.
 - B. Physically assemble the rocket and make iterative design adjustments.
 - C. Update the digital model based on results.
 - D. Repeat this cycle continuously.
- VIII. The rocket should be modular to allow for reuse and easy adaptation to changing competition requirements.
- IX. Design should prioritize aerodynamics and minimal weight.
- X. Conduct in-depth research on designs used by successful teams to inform decisions.

Specific Design Features

- XI. **Structural Rods:** Use carbon fiber or similar rods only in sections that do not house fragile payloads (e.g., eggs), as they have proven to be a risk in prior flights.
- XII. **Crew Stage Tube:** The diameter of the payload bay should be no less than **7 cm**, with **8–9 cm** preferred for optimal horizontal padding. Vertical padding is easier to implement; therefore, horizontal impact mitigation should guide design decisions.
- XIII. **Air Brake Design:** If an air brake is used, it should incorporate lightweight ball bearings to minimize friction. This allows for smaller servos and batteries, reducing overall system weight.
- XIV. **Fin Design:** Ensure the fins are sufficiently large enough to keep the rocket stable during flight.

Overall Engineering Approach

- XV. Prioritize *air brake-based apogee control* over initial mass adjustment. This method offers more consistent results across flights.
- XVI. Develop a *wind and atmospheric modeling system* based on local environmental conditions:
 - A. Use a drone outfitted with sensors (e.g., pitot tube, humidity, air density) to collect data at different altitudes.
 - B. Compare this data with open-source models (e.g., Open-Meteo, NOAA, NASA) using geographic coordinates.
 - C. Perform this data collection in multiple environments and weather conditions to generate a large dataset.
 - D. Identify correlations and build a predictive model with estimated error margins; this research may also support the development of a publishable paper.
- XVII. Use *CFD (Computational Fluid Dynamics)* to:
 - A. Determine drag coefficients (Cd) at various air brake extension lengths.
 - B. Model air pressure fluctuations to inform ideal part placement.
- XVIII. Build a *Python-based simulation tool* that:

- A. Integrates live or predicted atmospheric data (processed through the in-house model).
 - B. Includes parameters such as initial mass, moments of inertia, stability margins, and weathercocking.
 - C. Calculates the optimal air brake deployment height.
 - D. This system can be built as an extension of [EulerSim^{BETA}](#) for efficiency.
- XIX. Develop an onboard avionics system that autonomously deploys the air brake based on simulation output.
- XX. Conduct at least eight test flights to validate the system and refine deployment accuracy.